Modelling a 3D Printer

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https://github.com/ianpetek/MDS_mini_project

1 Physical Modelling

1.1 Kinematics

The first task is to derive the direct kinematics of the Core XY mechanism. This is slightly complicated by the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\Delta L = \Delta x + \Delta y, \qquad \Delta R = \Delta x - \Delta y, \qquad \Delta x = \frac{\Delta L + \Delta R}{2}, \qquad \Delta y = \frac{\Delta L - \Delta R}{2} \tag{1.1}$$

With ΔL and ΔR being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\Delta L = r_m \, \theta_L, \qquad \Delta R = r_m \, \theta_R \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the tool head is at $(0, 0)$, we get the following equations:

$$x(\theta_L, \theta_R) = \frac{r_m\theta_L + r_m\theta_R}{2} = \frac{r_m}{2}\left(\theta_L + \theta_R\right), \qquad y(\theta_L, \theta_R) = \frac{r_m\theta_L - r_m\theta_R}{2} = \frac{r_m}{2}\left(\theta_L - \theta_R\right) \tag{1.3}$$

$$\dot{x}(\dot{\theta}_L, \dot{\theta}_R) = \frac{r_m}{2}\left(\dot{\theta}_L + \dot{\theta}_R\right), \qquad \dot{y}(\dot{\theta}_L, \dot{\theta}_R) = \frac{r_m}{2}\left(\dot{\theta}_L - \dot{\theta}_R\right) \tag{1.4}$$

1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler–Lagrange equation. Additionally, we will split the problem into a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalised coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

1.2.1 Mechanical Part

Kinetic Energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{\text{mech}} = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there are no potential energy terms as it will be constant.

Dissipative Energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

$$\theta_{b,L1} = \frac{x + y}{r_b}, \qquad \theta_{b,R1} = \frac{x - y}{r_b} \tag{1.8}$$

These are used for the two fixed bearings on each belt. Then, for each side we get rotational dissipative energy:

$$D_{L1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_{R1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

For the two carriage bearings, if we assume the belt to be horizontal on the carriage like Figure 17 shows them, they are rotating only when the carriage moves along the x axis.

$$\theta_{b,L2} = \frac{x}{r_b}, \qquad \theta_{b,R2} = \frac{x}{r_b} \tag{1.9}$$

With that we get two more terms:

$$D_{L2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2, \qquad D_{R2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = \frac{\mu_r}{r_b^2} \left[4\dot{x}^2 + 2\dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2} \mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2} \mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_{\text{mech}} = \left[4 \frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[2 \frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.10}$$

1.2.2 Electrical Part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same.

Kinetic Energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly to (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{\text{el}} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2\right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2\right) \tag{1.11}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

Dissipative Energy

The only dissipative term in the electrical part of the system is the resistor:

$$D_{\text{el},L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{\text{el},R} = \tfrac{1}{2} i_R^2 R_m, \qquad D_{\text{el}} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2\right) \tag{1.12}$$

Matrix Equations

Finally, we realise that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\,\dot{q}^\top \mathbf{M}\,\dot{q}, \qquad D = \tfrac{1}{2}\,\dot{q}^\top \mathbf{N}\,\dot{q} \tag{1.13}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \dfrac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \dfrac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix} 8\dfrac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 4\dfrac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.14}$$

1.2.3 Generalised Force

Mechanical Coordinates

The non-conservative forces in the mechanical part are given by the motor torques τ_L and τ_R. The virtual work they perform is:

$$\Delta W = \tau_L\,\Delta\theta_L + \tau_R\,\Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius r_m, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m}\,\Delta x + \frac{\tau_L - \tau_R}{r_m}\,\Delta y$$

Applying the motor law $\tau = K_m i$, the generalised forces for x and y are:

$$Q_x = \frac{K_m}{r_m}\,(i_L + i_R)\,, \qquad Q_y = \frac{K_m}{r_m}\,(i_L - i_R) \tag{1.15}$$

Electrical Coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left(u_{m,L} - K_m\dot{\theta}_L\right)\Delta q_L + \left(u_{m,R} - K_m\dot{\theta}_R\right)\Delta q_R$$

Using the same transformation, we get the generalised forces for q_L and q_R:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m}\,(\dot{x} + \dot{y})\,, \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m}\,(\dot{x} - \dot{y}) \tag{1.16}$$

Lagrange Equations

Using the fact that this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2}\,\dot{q}^\top \mathbf{M}\,\dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2}\,\dot{q}^\top \mathbf{M}\,\dot{q}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt}\,(\mathbf{M}\dot{q}) = \mathbf{M}\ddot{q}, \qquad \frac{\partial \mathcal{L}}{\partial q} = 0, \qquad \frac{\partial D}{\partial \dot{q}} = \mathbf{N}\dot{q}$$

And finally solving the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.17}$$

Expanding back out of matrix form we get:

$$\left(m + J_m \frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m} i_L - \frac{K_m}{r_m} i_R = 0$$
$$\left(m + J_m \frac{2}{r_m^2}\right)\ddot{y} + \left(4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m} i_L + \frac{K_m}{r_m} i_R = 0$$
$$L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} = 0$$
$$L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} = 0$$

Or with substitutions, we get the final form of the four differential equations:

$$m_{\text{eq}} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} + \dot{y}\right) - \frac{R_m}{L_m} i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} - \dot{y}\right) - \frac{R_m}{L_m} i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.18}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \dfrac{di_L}{dt} \\ \dfrac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.19}$$

$$\mathbf{A} = \begin{bmatrix}
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & -\dfrac{b_x}{m_{\text{eq}}} & 0 & \dfrac{K_m}{m_{\text{eq}} r_m} & \dfrac{K_m}{m_{\text{eq}} r_m} \\
0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & -\dfrac{K_m}{m_{\text{eq}} r_m} \\
0 & 0 & -\dfrac{K_m}{r_m L_m} & -\dfrac{K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 \\
0 & 0 & -\dfrac{K_m}{r_m L_m} & \dfrac{K_m}{r_m L_m} & 0 & -\dfrac{R_m}{L_m}
\end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix}
0 & 0 \\
0 & 0 \\
0 & 0 \\
0 & 0 \\
\dfrac{1}{L_m} & 0 \\
0 & \dfrac{1}{L_m}
\end{bmatrix}$$

1.2.4 Decoupling the x and y Dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{array}{llll} i_x = i_L + i_R, & i_y = i_L - i_R, & u_{m,x} = u_{m,L} + u_{m,R}, & u_{m,y} = u_{m,L} - u_{m,R} \\ i_L = \dfrac{i_x + i_y}{2}, & i_R = \dfrac{i_x - i_y}{2}, & u_{m,L} = \dfrac{u_{m,x} + u_{m,y}}{2}, & u_{m,R} = \dfrac{u_{m,x} - u_{m,y}}{2} \end{array}} \tag{1.20}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m}i_x - \frac{R_m}{2L_m}i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m}i_x + \frac{R_m}{2L_m}i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\frac{di_x}{dt} = -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m}i_x + \frac{1}{L_m}u_{m,x}, \qquad \frac{di_y}{dt} = -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m}i_y + \frac{1}{L_m}u_{m,y} \tag{1.21}$$

And by substituting (1.20) into the first two equations of (1.18):

$$\ddot{x} = -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m}i_x, \qquad \ddot{y} = -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m}i_y \tag{1.22}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.23}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

1.3 Drawing Shapes

Rearranging the decoupled equations of motion (1.22), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{i_x = \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x}, \qquad i_y = \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y}} \tag{1.24}$$

The motor equations for each motor are:

$$u_{m,L} = L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\,(\dot{x} + \dot{y})$$
$$u_{m,R} = L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\,(\dot{x} - \dot{y})$$

Adding them, and using (1.20):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.21):

$$\boxed{u_{m,x} = L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\,\dot{x}, \qquad u_{m,y} = L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\,\dot{y}} \tag{1.25}$$

The original motor voltages then follow from (1.20):

$$u_{m,L} = \tfrac{1}{2}\,(u_{m,x} + u_{m,y})\,, \qquad u_{m,R} = \tfrac{1}{2}\,(u_{m,x} - u_{m,y}) \tag{1.26}$$

Since (1.25) contains $\frac{di_x}{dt}$, and (1.24) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any C^3 functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages are not realistic.

Ellipse

An ellipse with centre (h, k) and semi-axes a and b has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.27}$$

We also need the derivatives up to the third order:

$$\dot{x} = -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t, \qquad \dot{y} = b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t$$

Substituting into (1.24):

$$i_x(t) = -\frac{m_{\text{eq}} r_m}{K_m}\,a\cos t - \frac{b_x r_m}{K_m}\,a\sin t = -\frac{r_m}{K_m}\,a\,[m_{\text{eq}}\cos t + b_x \sin t]$$
$$i_y(t) = \frac{r_m}{K_m}\,b\,[b_y \cos t - m_{\text{eq}}\sin t]$$

And into (1.25):

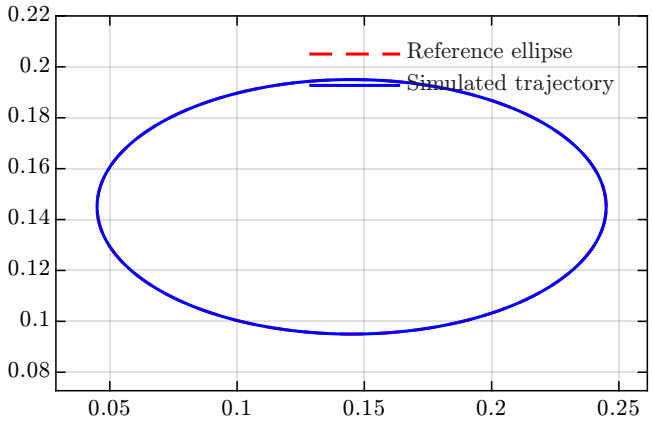
$$u_{m,x}(t) = L_m \frac{r_m}{K_m}\,a\,[m_{\text{eq}}\sin t - b_x\cos t] - R_m \frac{r_m}{K_m}\,a\,[m_{\text{eq}}\cos t + b_x \sin t] - \frac{2K_m}{r_m}\,a\sin t$$
$$u_{m,y}(t) = -L_m \frac{r_m}{K_m}\,b\,[m_{\text{eq}}\cos t + b_y \sin t] + R_m \frac{r_m}{K_m}\,b\,[b_y\cos t - m_{\text{eq}}\sin t] + \frac{2K_m}{r_m}\,b\cos t$$

With

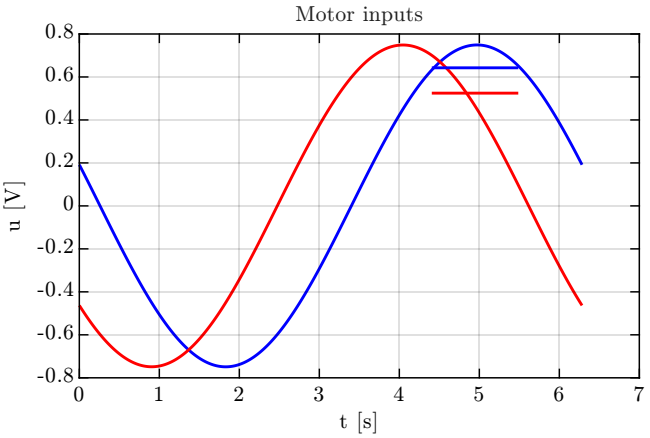
$$u_{m,L}(t) = \tfrac{1}{2}\,(u_{m,x}(t) + u_{m,y}(t))\,, \qquad u_{m,R}(t) = \tfrac{1}{2}\,(u_{m,x}(t) - u_{m,y}(t))$$

we obtain the analytical open-loop solution for drawing the ellipse.

Simulating the given system with parameters defined in Table 7, we get near-perfect alignment with the reference ellipse.



(a) Reference and simulated trajectory.

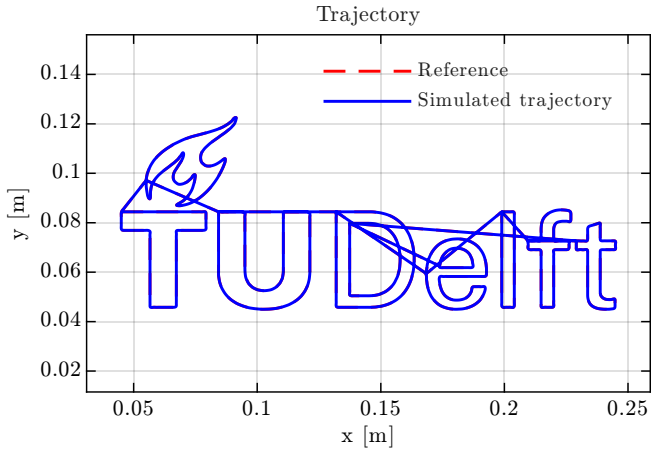


(b) Open-loop motor voltages.

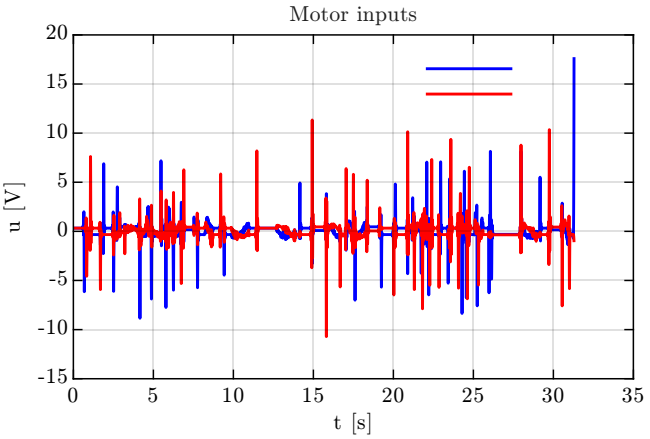
Figure 1: Open-loop drawing of an ellipse with nominal model parameters.

Arbitrary Drawing

As stated in the derivation, any C^3 function used for $x(t)$ and $y(t)$ will allow us to get an analytical solution. With that in mind we can draw arbitrary shapes by approximating them with quartic splines. Using MATLAB's *bwboundaries* function to detect edges of shapes, in combination with *spapi* which gives us an interpolating quartic spline given a set of points, we can draw the *TU Delft* logo[1].



(a) Reference and simulated trajectory.



(b) Open-loop motor voltages.

Figure 2: Open-loop drawing of the *TU Delft* logo with nominal model parameters.

1.3.1 Modelling the Nozzle

In the following section we model the nozzle of the 3D printer.

The system consists of an electrical and thermodynamical part. A circuit consisting of an inductor L_n, capacitor C_n and a heating resistor R_n. We choose the current through the inductor and the voltage across the capacitor, $i_{L,n}$ and u_c respectively, as first two states of our system. By solving the three Kirchhoff equations we get the following two first-order differential equations:

$$L_n \frac{di_{L,n}}{dt} = u_n - u_c, \qquad C_n \frac{du_c}{dt} = i_{L,n} - \frac{u_c}{R_n(T_r)} \tag{1.28}$$

Next, we model the thermal part. Besides the thermal capacitance of the nozzle $C_{t,n}$, given in the task, we also

[1]Different letter boundaries are connected by straight lines so the whole drawing can be one input function

model the thermal capacitance of the resistor C_r[2]. We choose two additional states, temperature of the resistor (T_r) and the temperature of the nozzle (T_n). Losses happening between the resistor and the nozzle are described by $R_{t,n}$, and the heat dissipation to the environment and to the filament is modelled with one variable R_{env}. The connection to the electrical part is the heat dissipated by the resistor $P_r = \frac{u_c^2}{R_n(T_r)}$. This gives us the following two equations which with (1.28) fully describe the system:

$$C_{t,n}\,\frac{dT_n}{dt} = \frac{T_r - T_n}{R_{t,n}} - \frac{T_n - T_e}{R_{\text{env}}}, \qquad C_r\,\frac{dT_r}{dt} = P_r - \frac{T_r - T_n}{R_{t,n}} \tag{1.29}$$

with

$$R_n(T_r) = R_0\left[1 + \alpha\,(T_r - T_0)\right] \tag{1.30}$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

Due to the square voltage term and the temperature dependence of the resistance the system is very non-linear, although the very different time constants of two systems (thermal states change much slower compared to the electrical) can be exploited. Alternatively, the model can reasonably be simplified by assuming a constant resistance. With that formulation we would get two linear systems connected by a single non-linear term.

1.3.2 Printing a circle of filament

In this section a 1D distributed parameter model for the cooling of a square cross-section filament on a heating plate and a numerical formulation of the problem are derived.

Modelling the system

We take a cross-section of $a \times a$ with length ΔL. Temperature is assumed uniform across the $a \times a$ cross-section to keep the model one dimensional in space. Three faces exchange heat with the environment (T_e), and one face with the heating plate (T_p).

The problem is modelled using the heat equation [2], Fourier's law gives the flux (1.31). Writing the energy density as (1.31) and substituting both into the energy balance yields the heat equation (1.32):

$$\partial_t u = -\nabla\cdot\phi + f, \qquad u = \rho c\,T, \qquad \phi = -\kappa\nabla T \tag{1.31}$$

$$\partial_t T = \frac{\kappa}{\rho c}\,\nabla^2 T + \frac{f}{\rho c} \tag{1.32}$$

Secondly, we need to describe the heat exchange of the segment ΔL. Using Newton's cooling law we can express the heat transfer through the surfaces as (1.33). Dividing that by V gives us the heat transfer per unit volume (1.34).

$$\Delta A_e = 3a\,\Delta L, \qquad \Delta A_p = a\,\Delta L, \qquad V = a^2\,\Delta L$$

$$\dot{Q} = K_e\,\Delta A_e\,(T_e - T) + K_p\,\Delta A_p\,(T_p - T) \tag{1.33}$$

$$f = \frac{\dot{Q}}{V} = \frac{1}{a}\Big[\,3K_e\,(T_e - T) + K_p\,(T_p - T)\,\Big] \tag{1.34}$$

Substituting everything back into (1.32), gives us the final PDE:

$$\boxed{\partial_t T = k\,\partial_{xx} T + \alpha\,T + \beta} \tag{1.35}$$

$$k = \frac{\kappa}{\rho c}, \qquad \alpha = -\frac{3K_e + K_p}{a\,\rho c}, \qquad \beta = \frac{3K_e T_e + K_p T_p}{a\,\rho c}$$

[2] If we do not model the thermal capacitance of the resistor, we end up with a differential-algebraic equation (DAE).

Front Fixing

Due to needing to solve the problem while the nozzle is still moving, this becomes a moving boundary problem. A method from [1] is used. We introduce a transformation ξ, that maps our space to the interval $[0,1]$, with $s(t)$ being the position of our boundary, and consequently $\dot{s}(t)$ the speed of the boundary/nozzle.

$$\xi = \frac{x}{s(t)}, \qquad 0 \le \xi \le 1, \qquad s(t) = s_0 + s_1 t, \qquad \dot{s} = s_1 \tag{1.36}$$

Applying a change of variable to our equation (1.35) we get:

$$\frac{\partial T}{\partial x} = \frac{1}{s}\frac{\partial T}{\partial \xi}, \qquad \frac{\partial^2 T}{\partial x^2} = \frac{1}{s^2}\frac{\partial^2 T}{\partial \xi^2}$$

$$\left(\frac{\partial T}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi + \frac{\partial T}{\partial \xi}\left(\frac{\partial \xi}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{x\,\dot{s}}{s^2}\frac{\partial T}{\partial \xi} = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi}$$

$$\boxed{\left(\frac{\partial T}{\partial t}\right)_\xi = \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi} + \frac{k}{s^2}\frac{\partial^2 T}{\partial \xi^2} + \alpha T + \beta} \tag{1.37}$$

Boundary and Initial Conditions

With $T(\xi,t)$ we can place necessary boundary conditions for the problem to be solvable. Firstly, the temperature at the endpoint ($\xi = 1$) at all times equals the temperature of the nozzle which constitutes a Dirichlet condition. Secondly, we assume that the heat dissipation on the ($\xi = 0$) surface is negligible compared to the whole surface, so a Neumann boundary condition is used to model it insulated. Finally, as seen in (1.36), there exists initial length of filament s_0, needed to solve the problem, and its temperature at $t = 0$ is assumed uniform and equal to the nozzle temperature.

$$T(1,t) = T_n, \qquad \partial_\xi T(0,t) = 0, \qquad T(\xi,0) = T_n, \qquad s_0 > 0 \tag{1.38}$$

Discretisation

The temperature is discretised on a uniform grid in the transformed coordinate $\xi \in [0,1]$, with spacing $\Delta\xi = 1/N$, and in time with step Δt. Time is advanced with an implicit (backward) Euler scheme, the diffusion term $\partial_{\xi\xi}T$ uses a second-order central difference, which is symmetric and produces a tridiagonal matrix and the advection term $\partial_\xi T$ uses a first-order upwind difference.

$$\xi_i = i\,\Delta\xi, \quad i = 0,\dots,N, \qquad t_j = j\,\Delta t, \qquad T_i^j \approx T(\xi_i, t_j)$$

$$\left.\frac{\partial T}{\partial t}\right|_i^j \approx \frac{T_i^j - T_i^{j-1}}{\Delta t}, \qquad \left.\frac{\partial T}{\partial \xi}\right|_i^j \approx \frac{T_{i+1}^j - T_i^j}{\Delta \xi}, \qquad \left.\frac{\partial^2 T}{\partial \xi^2}\right|_i^j \approx \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta \xi^2} \tag{1.39}$$

$$\frac{T_i^j - T_i^{j-1}}{\Delta t} = \frac{\xi_i\,\dot{s}}{s}\frac{T_{i+1}^j - T_i^j}{\Delta \xi} + \frac{k}{s^2}\frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta \xi^2} + \alpha T_i^j + \beta \tag{1.40}$$

With some substitutions and after inserting the approximations into the transformed PDE gives we get:

$$\gamma_i = \frac{\xi_i\,\dot{s}\,\Delta t}{s\,\Delta\xi} = \frac{i\,\dot{s}\,\Delta t}{s}, \qquad g = \frac{k\,\Delta t}{s^2\,\Delta\xi^2}, \qquad s,\ \dot{s} \text{ evaluated at } t_j$$

$$\boxed{T_i^j\left[1 + \gamma_i + 2g - \alpha\,\Delta t\right] = g\,T_{i-1}^j + (\gamma_i + g)\,T_{i+1}^j + T_i^{j-1} + \beta\,\Delta t} \tag{1.41}$$

We can see that we arrive at an implicit formulation, where we need to solve a linear system at each time step.

Linear System at $t = t_j$

Writing down a couple of terms we can see that we get a tridiagonal matrix, with the first row representing the Neumann condition and the last the Dirichlet condition.

$$d_i = 1 + \gamma_i + 2g - \alpha\,\Delta t, \qquad e_i = \gamma_i + g, \qquad T_0^j = T_1^j, \qquad T_N^j = T_n$$

$$\begin{bmatrix} 1 & -1 & & & & \\ -g & d_1 & -e_1 & & & \\ & -g & d_2 & -e_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -g & d_{N-1} & -e_{N-1} \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{N-1}^j \\ T_N^j \end{bmatrix} = \begin{bmatrix} 0 \\ T_1^{j-1} + \beta\,\Delta t \\ T_2^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-1}^{j-1} + \beta\,\Delta t \\ T_n \end{bmatrix} \tag{1.42}$$

The fact that the matrix is tridiagonal allows us to use specialised solvers like *scipy.linalg.solve_banded*.

After the Boundary Stops (Closed Ring)

After the boundary stops moving, i.e. the circle is closed the matrix needs to be slightly modified, which causes it to lose its tridiagonal form. At the time point of contact the first and last node are connected and become one node, and its temperature is set to the mean of both of them. As the boundary stops moving $\dot{s}$ and consequently γ_i terms disappear.

$$\dot{s} = 0 \;\Rightarrow\; \gamma_i = 0, \qquad s = L, \qquad T_N^j \equiv T_0^j, \qquad g = \frac{k\,\Delta t}{L^2\,\Delta\xi^2}, \qquad d = 1 + 2g - \alpha\,\Delta t$$

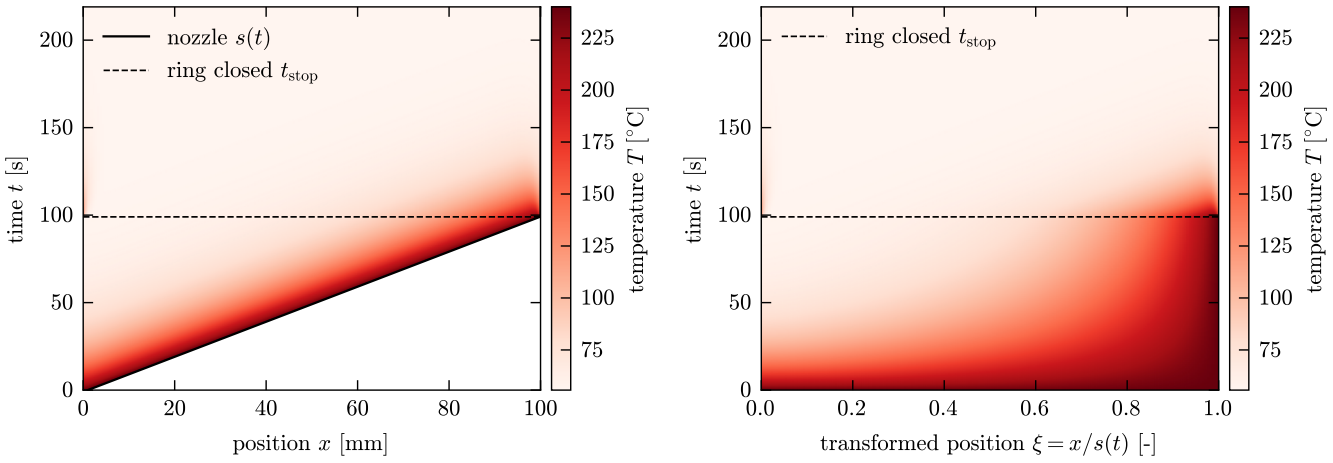
$$\begin{bmatrix} d & -g & & & -g \\ -g & d & -g & & \\ & \ddots & \ddots & \ddots & \\ & & -g & d & -g \\ -g & & & -g & d \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ \vdots \\ T_{N-2}^j \\ T_{N-1}^j \end{bmatrix} = \begin{bmatrix} T_0^{j-1} + \beta\,\Delta t \\ T_1^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-2}^{j-1} + \beta\,\Delta t \\ T_{N-1}^{j-1} + \beta\,\Delta t \end{bmatrix} \tag{1.43}$$

First step after connection:

$$T_0^{j-1} \leftarrow \tfrac{1}{2}\left(T_0^{j-1} + T_N^{j-1}\right)$$

Simulation

The derived model was simulated in *Python* with the parameters listed in Table 8 (Appendix C).



(a) Physical space x.

(b) Transformed space ξ.

Figure 3: Temperature of the filament while printing and after the ring is closed.

2 Finite State Machine

2.1 Filament Replacement

The filament replacement is modelled with two machines: a printing machine and a filament replacement machine.

2.1.1 Formal Machine Definition

The FSM is defined as the Mealy machine

$$\mathcal{M} = (Q, \Sigma, \Lambda, \delta, \lambda, q_0). \tag{2.1}$$

States. The state $q = (p, f, r)$ is the printing state, the filament replacement state and the retries left,

$$\begin{aligned} Q &= P \times \mathcal{F} \times R, \\ P &= \{\texttt{IDLE}, \texttt{IN_PROG}, \texttt{PAUSED}\}, \\ \mathcal{F} &= \{\texttt{IDLE}, \texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}, \\ R &= \{0, 1, 2, 3\}, \end{aligned} \tag{2.2}$$

where `MOVE` moves to the wipe location, `WIPE` wipes the excess filament, `COOL` switches the nozzle heat off, `LOAD` loads the new filament and `HEAT` switches the nozzle heat on. The head location is saved on entering `MOVE` and the print is resumed on leaving `HEAT`. The initial state is $q_0 = (\texttt{IDLE}, \texttt{IDLE}, 3)$.

Inputs. An input is a pair $\sigma = (e, y)$. Here e is the external event (C, F or nothing) and y is the feedback of the current step: ok_k or fail_k for each step $k \in K = \{\texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}$, or nothing. This gives $3 \cdot 11 = 33$ inputs:

$$\begin{aligned} E &= \{C,\ F,\ \varepsilon\}, \\ Y &= \{\varepsilon,\ \text{ok}_{\texttt{MOVE}},\ \text{fail}_{\texttt{MOVE}},\ \text{ok}_{\texttt{WIPE}},\ \text{fail}_{\texttt{WIPE}},\ \ldots,\ \text{ok}_{\texttt{HEAT}},\ \text{fail}_{\texttt{HEAT}}\}, \\ \Sigma &= E \times Y. \end{aligned} \tag{2.3}$$

Outputs. The outputs Λ are messages to the user, for example "filament change in progress", "k successful", "k failure, trying again", "filament change process failed, initiate filament change again" and "filament change already in progress".

Transition and output functions. The transition function $\delta : Q \times \Sigma \to Q$ and output function $\lambda : Q \times \Sigma \to \Lambda$ are given by Table 1 and Figures 4 and 5.

Implementation. The machine is a transition table (Table 1) in `fsm.ipynb`. The rows matching the current state and input are checked and the corresponding outputs are determined. `X` means `DON'T CARE`.

Table 1: Snapshot of the state transition table (state names abbreviated, `X` is a `DON'T CARE`, empty cells mean no signal).

p	f	e	y	r	p'	f'	message	r'
IDLE	IDLE	C		X	IDLE	MOVE	filament change in progress	3
IN_PROG	IDLE	C		X	PAUSED	MOVE	filament change in progress	3
X	MOVE		ok_MOVE	X	X	WIPE	moving to wiping location successful	3
X	MOVE		fail_MOVE	3	X	MOVE	failure, trying again	2
X	MOVE		fail_MOVE	0	X	IDLE	process failed, initiate again	3
PAUSED	HEAT		ok_HEAT	X	IN_PROG	IDLE	nozzle heating successful	3
X	WIPE	C	X	X	X	WIPE	filament change already in progress	X

The model was checked against a set of different test cases (`test_cases.csv`).

2.1.2 Transition

The two sub-machines are shown in Figures 4 and 5. Transitions are labelled *input* [*guard*] / *action*, where r is the number of retries left. A C or F event starts a replacement only when the filament machine is idle; otherwise it is ignored. If it arrives together with a feedback signal, the event takes precedence.

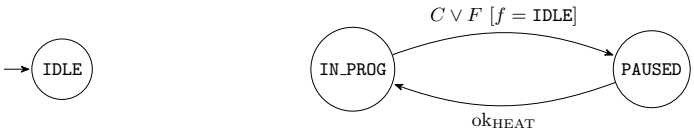


Figure 4: Printing sub-machine with state $p \in P$. The transition from IDLE to IN_PROG (print started) belongs to the print job handling and is not modelled here.

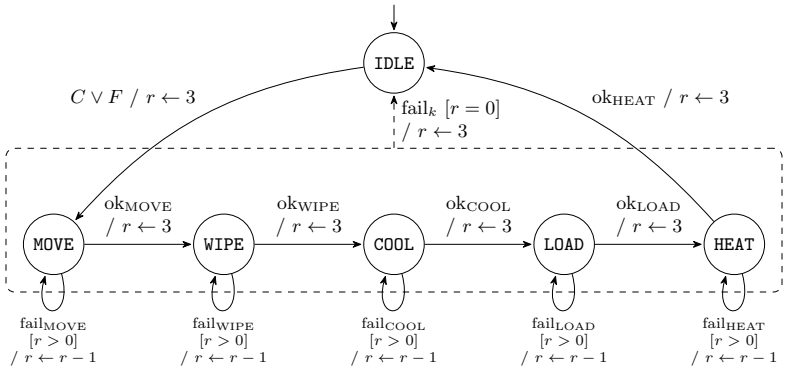


Figure 5: Filament replacement sub-machine with state $f \in \mathcal{F}$. For readability the five fail$_k$ [$r = 0$] transitions from the steps k back to IDLE are drawn as a single dashed arrow from the box around the steps. The two sub-machines are synchronised: the $C \vee F$ event and the ok$_{\text{HEAT}}$ event act on both machines in the same step.

2.2 Parked Detection

The head is parked at the origin when both motions have hit their end of line, i.e. a motion was commanded but the motor did not rotate. This is detected by the DFA

$$\mathcal{A} = (Q_A, \Sigma_A, \delta_A, q_{A,0}, F_A) \tag{2.4}$$

with

$$\begin{aligned} Q_A &= \{q_0, q_1, q_2, q_3\}, \\ \Sigma_A &= M \times m = \{(H,+), (H,0), (V,+), (V,0)\}, \\ q_{A,0} &= q_0, \qquad F_A = \{q_3\}, \end{aligned} \tag{2.5}$$

where the states have the meaning

- q_0: no end of line detected,
- q_1: horizontal end of line (left side) detected,
- q_2: vertical end of line (bottom side) detected,
- q_3: both detected, the head is parked.

The transition function $\delta_A : Q_A \times \Sigma_A \to Q_A$ is given in Table 2 and drawn in Figure 6.

Table 2: Transition function δ_A of the parked detector.

q	$(H,+)$	$(H,0)$	$(V,+)$	$(V,0)$
$\rightarrow q_0$	q_0	q_1	q_0	q_2
q_1	q_0	q_1	q_1	q_3
q_2	q_2	q_3	q_0	q_2
$*\, q_3$	q_2	q_3	q_1	q_3

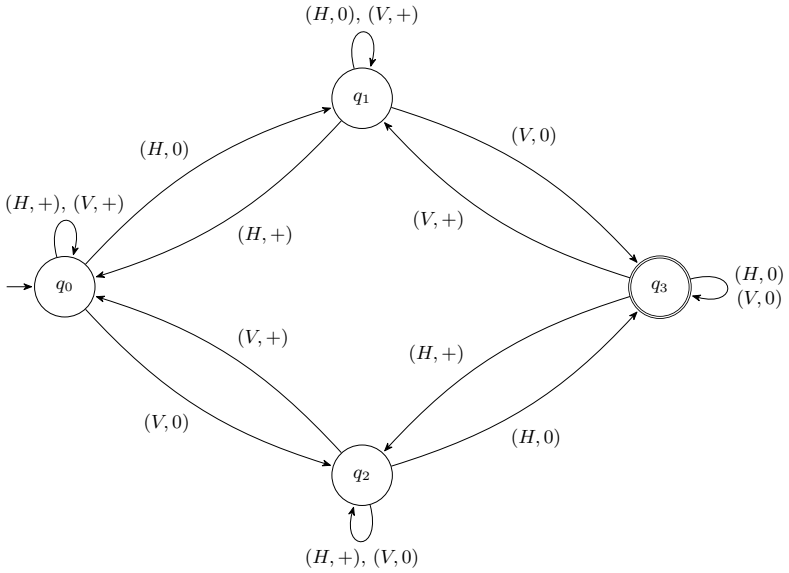


Figure 6: Transition of the parked detector $\mathcal{A}$. The accepted language is the set of sequences σ that end in q_3.

2.3 Detecting the Other Corners

The automaton is extended to the motions H_x, H_y and V and to the angular velocities $+$, $-$ and 0,

$$\begin{aligned} \mathcal{A}' &= (Q_A, \Sigma'_A, \delta'_A, q_0, F_A), \\ \Sigma'_A &= \{H_x, H_y, V\} \times \{+, -, 0\}, \end{aligned} \tag{2.6}$$

with Q_A, q_0 and F_A as before. The transition function $\delta'_A : Q_A \times \Sigma'_A \rightarrow Q_A$ is given in Table 3 and drawn in Figure 7.

Table 3: Transition function δ'_A of the corner detector.

q	$(H_x,\pm)$	$(H_x,0)$	$(H_y,\pm)$	$(H_y,0)$	$(V,\cdot)$
$\rightarrow q_0$	q_0	q_1	q_0	q_2	q_0
q_1	q_0	q_1	q_1	q_3	q_1
q_2	q_2	q_3	q_0	q_2	q_2
$*\, q_3$	q_2	q_3	q_1	q_3	q_3

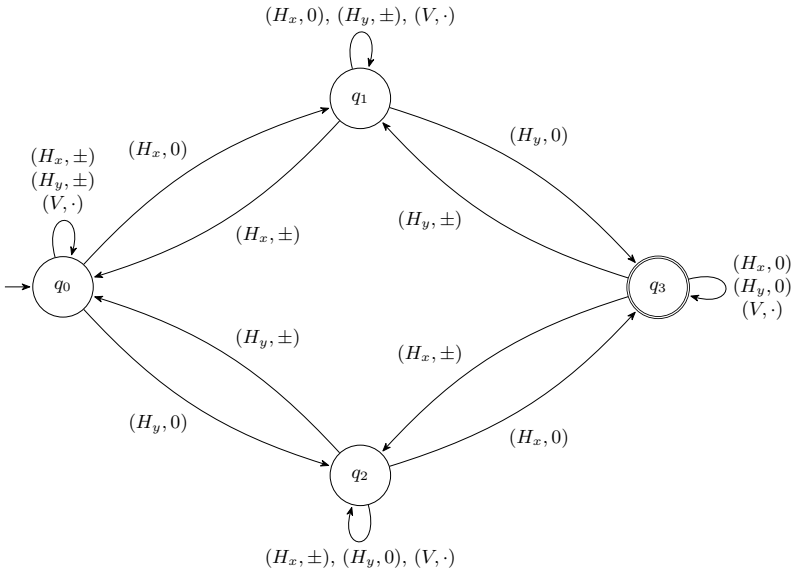


Figure 7: Transition of the corner detector $\mathcal{A}'$.

3 Hybrid System

3.1 Part 1: Formal Definition

We define the hybrid state machine as

$$H = (Q, X, U, Y, f, h, Init, E, G, R).$$

It combines three smaller machines running in parallel: the nozzle machine $\mathcal{N}$, the printer machine $\mathcal{P}$ and the filament machine $\mathcal{M}$.

In mode q the continuous state flows as $\dot{x} = f(q, x, u)$. An edge is taken as soon as its guard is true, and z^+ denotes the value of z right after the transition. There are no separate invariants.

Q: set of modes. One mode from each machine, $Q = Q_{\mathcal{N}} \times Q_{\mathcal{P}} \times Q_{\mathcal{M}}$:

$$\begin{aligned} Q_{\mathcal{N}} &= \{\text{COOL}, \text{HEATING_UP}, \text{HEATED}, \text{COOLING}\}, \\ Q_{\mathcal{P}} &= \{\text{IDLE}, \text{WAIT_HOT}, \text{TRAVEL_TO_START}, \text{PRINT_PATH}, \\ &\qquad \text{PAUSED}, \text{COMPLETED}\}, \\ Q_{\mathcal{M}} &= \{\text{IDLE}, \text{MOVING_TO_WIPE}, \text{WIPING}, \text{NOZZLE_COOLING}, \\ &\qquad \text{LOADING_FILAMENT}, \text{NOZZLE_HEATING}, \text{READY_TO_RESUME}\}. \end{aligned}$$

WAIT_HOT, TRAVEL_TO_START and PRINT_PATH together form the mode IN_PROGRESS of $\mathcal{P}$.

X: continuous state. The printer state $x_p = (x, y, \dot{x}, \dot{y}, T_n, T_r)$ (head position and velocity, nozzle and heater temperatures) plus the variables of Table 4. We use the reduced plant, where the fast electrical dynamics are instantaneous ($L_m \approx 0$, $u_c = u_n$).

Table 4: Variables in X besides the printer state x_p.

Symbol	Meaning
L	Length of filament used so far
τ	Time spent in the current mode; a timer with $\dot{\tau} = 1$ that is reset to 0 on every edge
k	Index of the current point in the path, which is an array of coordinates that the printer follows; $k \in \{1, \dots, n\}$
(x_s, y_s)	Coordinates of the head saved when the printer is paused
req, *heated*	Request to the nozzle to start heating; indicator that the nozzle is heated
paused	Indicator that the printer is in the paused state
change_active, *change_done*	Indicator that a filament change is in progress; indicator that a filament change has finished

$U = U_D \times U_C$: inputs. $U_D = \{C, START\}$ are the user events (colour change, start). The continuous inputs U_C are listed in Table 5. They are fixed by the mode, as given in Section 3.2. The motors are driven by $\mathcal{M}$ when it is not IDLE and by $\mathcal{P}$ otherwise.

Table 5: Continuous inputs U_C.

Symbol	Code variable	Meaning
(u_{mL}, u_{mR}, u_n)	`mode_motor`, `mode_nozzle`	Left and right motor voltages and heater voltage; each is either off or on, depending on the mode
Extruder switch	`is_extruding`	Whether the extruder is on
$r = (x_c, y_c)$	`command_x`, `command_y`	Commanded head position that the motors drive towards

Y: outputs. The outputs are the head position (x, y), the nozzle temperature T_n, the length of filament used L, and the current mode of each machine $(q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$. Here the mode of a machine is the step it is currently in, for example HEATED for $\mathcal{N}$, PAUSED for $\mathcal{P}$ or WIPING for $\mathcal{M}$.

f: flow map. The reduced CoreXY plant (`sys_reduced`): a damped head mass driven by the two coupled motors, and a two-node nozzle thermal model heated with power $u_n^2/R_n(T_r)$. The other variables follow Table 4.

h: output map. h computes the outputs from the current mode $q = (q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$ and continuous state x, so $y_{out} = h(q, x)$. It does not change the values; it only selects which quantities are observed:

$$y_{out} = h(q, x) = \big(\, x,\ y,\ T_n,\ L,\ q_{\mathcal{N}},\ q_{\mathcal{P}},\ q_{\mathcal{M}} \,\big).$$

The first four entries are the head position (x, y), T_n and L, read directly from x_p and L. The last three come from q. The remaining variables ($\dot{x}$, $\dot{y}$, T_r, τ, k, the saved coordinates and the flags) are internal and not part of the output.

***Init*: initial state.** $Init = \{(q_0, x_0)\}$ contains a single element:

$$\begin{aligned} q_0 &= (\text{COOL}, \text{IDLE}, \text{IDLE}), \\ x_p(0) &= (x, y, \dot{x}, \dot{y}, T_n, T_r) = (0, 0, 0, 0, T_e, T_e), \\ (L, \tau, k) &= (0, 0, 1), \\ (\mathit{req}, \mathit{heated}, \mathit{paused}, \mathit{change_active}, \mathit{change_done}) &= (\text{false}, \text{false}, \text{false}, \text{false}, \text{false}), \end{aligned}$$

where T_e is the ambient temperature.

Shorthand for the guards. The conditions in Table 6 are used to keep the guards short.

Table 6: Shorthand used in the guards.

Symbol	Meaning
arr	Head has arrived at its target: $\lvert x - x_c\rvert < \varepsilon$ and $\lvert y - y_c\rvert < \varepsilon$, with $\varepsilon = 0.5\,\mathrm{mm}$
F	Spool is empty: $L \geq L_{spool}$
T_{hi}	Temperature at which the nozzle counts as hot, $T_{hi} = 350\,\mathrm{K}$
$arr_{>0.5}$	arr has held for more than $0.5\,\mathrm{s}$
$(T_n \geq T_{hi})_{>1}$	$T_n \geq T_{hi}$ has held for more than $1\,\mathrm{s}$
$C^{\uparrow}$	Button C has just been pressed (rising edge, $0 \to 1$)

E: edges. $E \subseteq Q \times Q$ is the set of allowed transitions $e = (q, q')$. It is the union of the edges of the three machines, which are listed in Section 3.2:

$$E = E_{\mathcal{N}} \cup E_{\mathcal{P}} \cup E_{\mathcal{M}},$$
$$e = (q, q'), \quad \text{e.g. } e = (\mathrm{COOL}, \mathrm{HEATING_UP}) \in E_{\mathcal{N}}.$$

G: guards. $G(e)$ is the condition on x and u under which the edge e is taken:

$$G(\mathrm{COOL}, \mathrm{HEATING_UP}) = req,$$
$$G(\mathrm{HEATING_UP}, \mathrm{HEATED}) = (T_n \geq T_{hi})_{>1}.$$

The edges IN_PROGRESS $\to$ PAUSED in $\mathcal{P}$ and IDLE $\to$ MOVING_TO_WIPE in $\mathcal{M}$ share the guard $C^{\uparrow} \vee F$, so they fire together.

R: resets. $R(e)$ gives the values z^+ of the variables right after e is taken:

$$R(\mathrm{IDLE}, \mathrm{WAIT_HOT}): \; req^+ = \mathrm{true},$$
$$R(\mathrm{PRINT_PATH}, \mathrm{PRINT_PATH}): \; k^+ = k + 1.$$

Every edge also resets $\tau^+ = 0$ (not repeated in the tables), and any variable not listed keeps its value, $z^+ = z$.

3.2 Part 2: The Machines

Each machine is shown as a flowchart that gives its modes, the inputs u used in each mode, and the edges with their guard G and reset R, followed by a short explanation.

3.2.1 Nozzle Machine $\mathcal{N}$

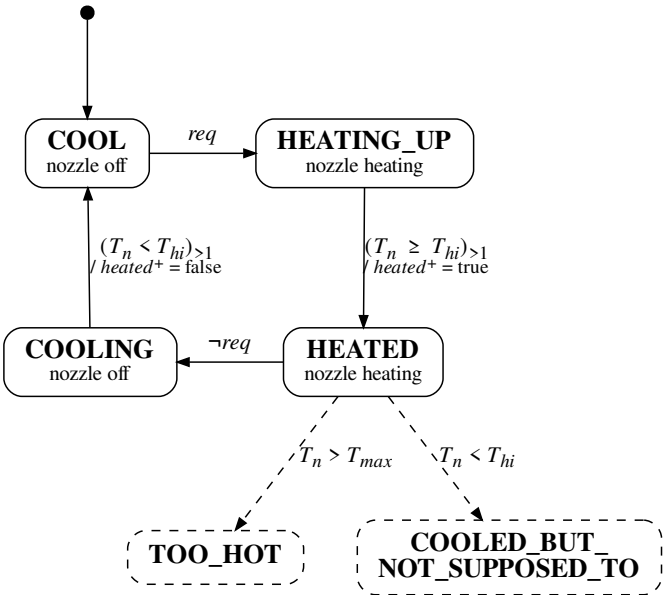


Figure 8: Nozzle machine, in the notation of this report. The dashed modes are failure modes of the implementation, which are not part of H. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- The nozzle starts in COOL with the heater off. When heating is requested (*req*), it moves to HEATING_UP and heats.
- After $T_n \geq T_{hi}$ has held for more than 1 s it is HEATED and *heated* becomes true. The machine keeps heating and does not regulate T_n further.
- When *req* is withdrawn the heater is switched off (COOLING), and after $T_n < T_{hi}$ for more than 1 s it returns to COOL with *heated* false.

3.2.2 Printer Machine $\mathcal{P}$

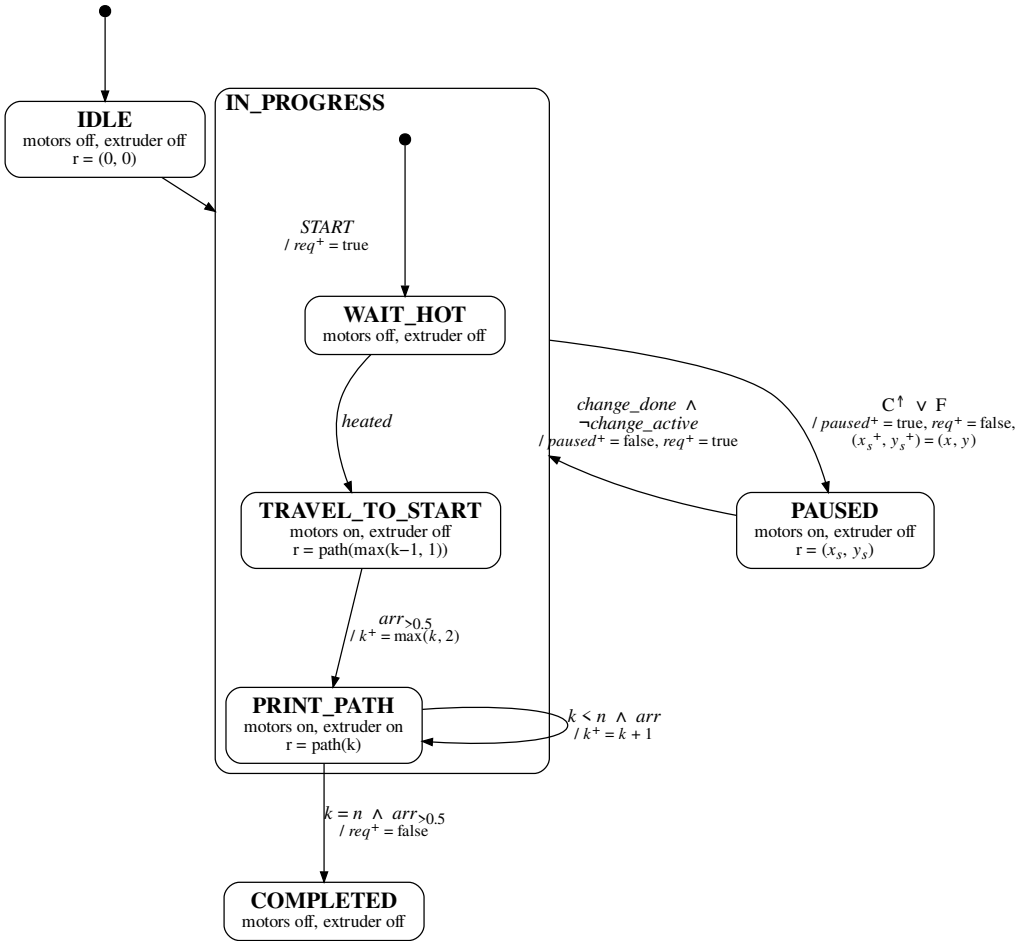


Figure 9: Printer machine, in the notation of this report. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- *START* requests heating and the printer waits in WAIT_HOT until the nozzle is *heated*.
- The head then travels to the start of the path and prints it point by point, increasing k each time it arrives. After the last point the printer is COMPLETED.
- On $C^{\uparrow}$ or F the printer pauses from any sub-mode of IN_PROGRESS and saves the head position (x_s, y_s). When the change is done it re-enters WAIT_HOT, so it waits for the nozzle, travels back to path$(k-1)$ and continues from point k.

3.2.3 Filament Machine $\mathcal{M}$

The wipe location is (x_w, y_w), the wipe time is t_w and the load time is t_l.

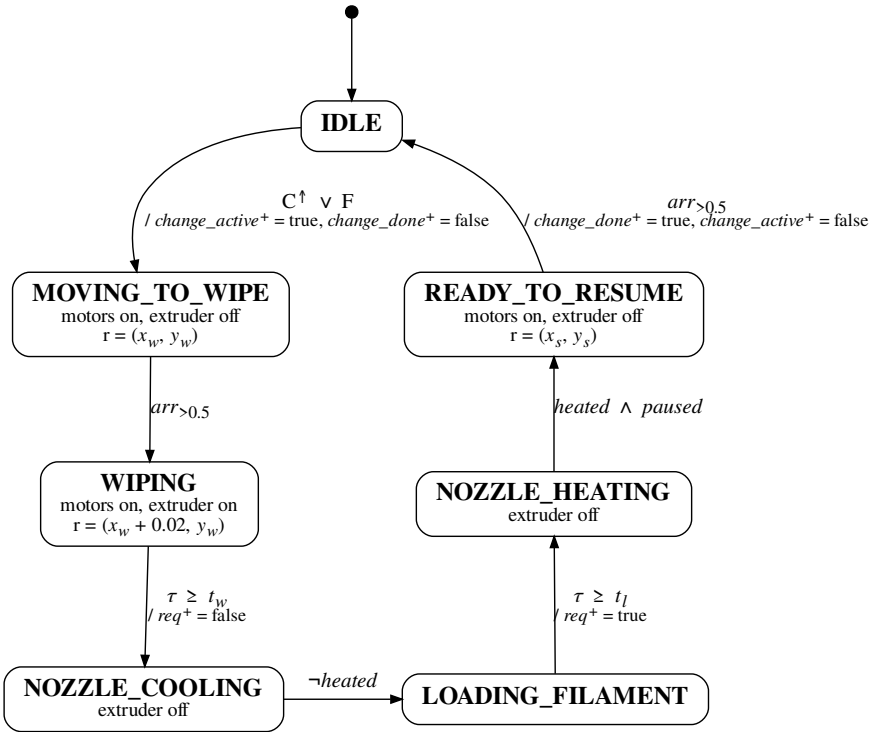


Figure 10: Filament machine, in the notation of this report. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- On $C^{\uparrow}$ or F the change starts: the head moves to the wipe location (x_w, y_w) and wipes with the extruder on for $t_w = 3\,\mathrm{s}$.
- Heating is then switched off (*req* false). Once the nozzle has cooled ($\neg$ *heated*), new filament is loaded for $t_l = 5\,\mathrm{s}$.
- Heating is requested again. When the nozzle is *heated* and the printer is paused, the head returns to (x_s, y_s) and *change_done* becomes true, which releases the printer.

Note. The nozzle flowchart shows two failure modes of the implementation (dashed). The implementation also has a *FAILED* mode and a manual pause and resume input. The definitions above ignore these failure modes and all retries, and cover only the successful filament change.

3.3 Part 3: Drawing a Square

Path Tracking The square is drawn from a list of points $\mathrm{path}(k) = (x_k, y_k)$, $k = 1, \ldots, n$, with $n = 25$.

- The path runs counter-clockwise through the corners $(0.05, 0.05)$, $(0.25, 0.05)$, $(0.25, 0.25)$, $(0.05, 0.25)$ (in metres) and back to the start.
- Each side has five intermediate points, so consecutive points are 33.3 mm apart.
- In PRINT_PATH the head is commanded to $\mathrm{path}(k)$ with the extruder on.
- Once *arr* holds, the index is increased to $k + 1$ and the head moves to the next point.
- The square is therefore drawn point by point, until $k = n$.

Test case. A single print covers both a filament run-out F and a colour change C. In every filament change the head moves to the wipe location $(0, 0.5)\,\mathrm{m}$.

- Printing starts at $t \approx 1\,\mathrm{s}$ with the first spool (red).
- The spool runs out (F) at $t \approx 75.0\,\mathrm{s}$. The printer pauses, the filament is changed, and printing resumes at $t \approx 101.3\,\mathrm{s}$ with the new spool (orange).

- The new colour is not wanted, so C is pressed at $t \approx 129.3\,\text{s}$. A second filament change runs and printing resumes at $t \approx 155.8\,\text{s}$ (green).
- The square is complete at $t \approx 188.8\,\text{s}$.

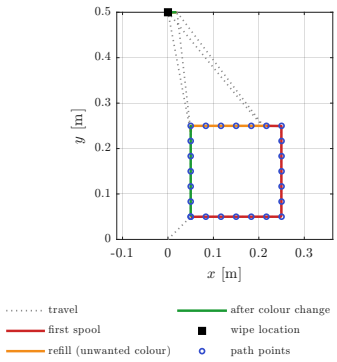


Figure 11: Head path: first spool (red), unwanted colour (orange), after the colour change (green) and travel moves.

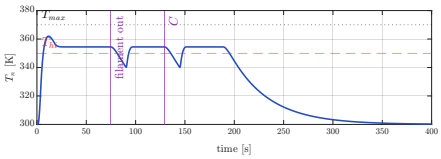


Figure 12: Nozzle temperature T_n with the thresholds T_{hi} and T_{max}.

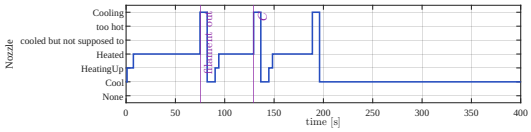


Figure 13: Nozzle machine mode.

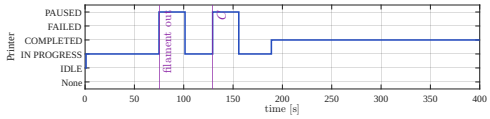


Figure 14: Printer machine mode.

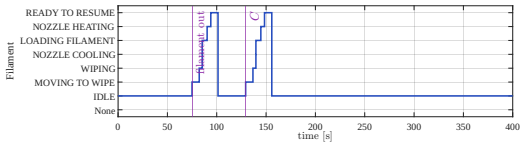


Figure 15: Filament machine mode.

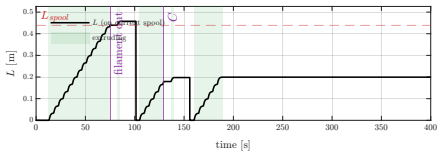


Figure 16: Extruded filament length L against L_{spool}.

References

[1] John Crank. *Free and Moving Boundary Problems*. Clarendon Press, Oxford, 1984.

[2] Ton J. J. van den Boom, Manuel Mazo Jr., and Riccardo M. G. Ferrari. Modeling of dynamical systems. Lecture notes for the course SC42155 (draft), Delft Center for Systems and Control, Delft University of Technology, September 2026.

A Symbol Table

Table 7: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_L	–
Motor voltage left	$u_{m,L}$	–
Motor current left	i_L	–
Motor radius	r_m	11 mm
Motor inductance	L_m	10 mH
Motor resistance	R_m	90 Ω
Motor inertia	J_m	8×10^{-6} kg m^2
Motor constant	K_m	0.07 N m/A
Tool head position	x, y	–
Tool head mass	m	60 g
Nozzle voltage	u_n	–
Nozzle inductor current	$i_{L,n}$	–
Nozzle capacitance	C_n	–
Nozzle inductance	L_n	–
Capacitor voltage	u_c	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	9 K/W
Nozzle resistance	R_n	8 Ω
Resistor thermal capacity	C_r	–
Resistor temperature	T_r	–
Thermal resistance to environment	R_{env}	–
Reference resistance	R_0	–
Reference temperature	T_0	–
Temperature coefficient of resistance	α	–
Nozzle temperature	T_n	240 °C
Environmental temperature	T_e	26.85 °C
Bearing friction rolling	μ_r	2.7×10^{-8} N m s/rad
Bearing friction linear x-direction	$\mu_{l,x}$	0.002 N s/m
Bearing friction linear y-direction	$\mu_{l,y}$	0.001 N s/m
Bearing radius	r_b	3 mm
Print bed temperature	T_b	105 °C
Print area dimension x	d_x	29 cm
Print area dimension y	d_y	29 cm

B Core XY Mechanism

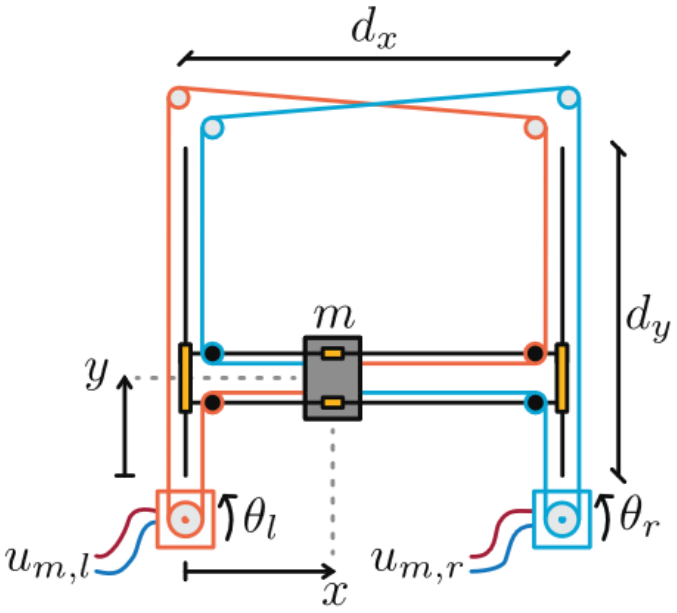


Figure 17: Schematic of the Core XY mechanism.

C Filament Simulation Parameters

Table 8: Parameters of the filament cooling simulation.

Description	Symbol	Value
Density	ρ	$1240\,\mathrm{kg/m^3}$
Specific heat	c	$1800\,\mathrm{J/(kg\,K)}$
Thermal conductivity	κ	$0.13\,\mathrm{W/(m\,K)}$
Cross-section side	a	$2\,\mathrm{mm}$
Heat transfer coefficient, environment	K_e	$10\,\mathrm{W/(m^2\,K)}$
Heat transfer coefficient, heating plate	K_p	$200\,\mathrm{W/(m^2\,K)}$
Environmental temperature	T_e	$26.85\,^\circ\mathrm{C}$
Heating plate temperature	T_p	$60\,^\circ\mathrm{C}$
Initial filament length	s_0	$1\,\mathrm{mm}$
Nozzle speed	s_1	$1\,\mathrm{mm/s}$
Ring length	L	$100\,\mathrm{mm}$
Nozzle temperature	T_n	$240\,^\circ\mathrm{C}$
Number of intervals	N	200
Time step	Δt	$0.05\,\mathrm{s}$
Simulated time after closing	–	$120\,\mathrm{s}$

AI Acknowledgment

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- explaining concepts;
- spell and grammar checking;
- generating plots in the same style;
- transcribing written equations from paper to LaTeX;
- assistance with syntax for MATLAB and Python;
- creating TikZ diagrams from Stateflow charts.